

Exogenic τ -Cascade in Emergent Necessity Theory: Definition, Cross-Domain Taxonomy, and Falsifiability

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Abstract

We formalize the notion of an **Exogenic τ -Cascade** within Emergent Necessity Theory (ENT). An exogenic τ -Cascade is defined as an external perturbation $F_k(t)$ that couples into a domain i and induces recursive drift or coherence shifts, thereby altering $\tau_i = E_{\text{syn},i}/E_{\text{drift},i}$ or its effective pressure $\kappa_{R,i}^{\text{eff}}$. We show how such cascades can, in principle, propagate across domains (physical $\rightarrow$ atmospheric $\rightarrow$ biospheric $\rightarrow$ symbolic). The framework is source-agnostic, applying equally to celestial flybys, solar storms, volcanic aerosols, anthropogenic shocks, and informational floods. We provide a taxonomy of cascade classes, explicit falsifiability criteria (C1C3), and a minimal simulation protocol. The theory is structured to ensure empirical testability: many hypothesized cascades will produce null results, which in ENT strengthens bounds on cross-domain couplings rather than weakening the framework.

1 Core ENT Quantities

For each domain $i \in \mathcal{D}$ we define:

$$\tau_i(t) = \frac{E_{\text{syn},i}(t)}{E_{\text{drift},i}(t)}, \quad \tau_{c,i} > 0,$$

where $E_{\text{syn},i}$ sustains recursive coherence, $E_{\text{drift},i}$ injects noise/disruption, and $\tau_{c,i}$ is the critical threshold.

The recursive pressure is:

$$\kappa_{R,i}(t) = \frac{\dot{\tau}_i(t)}{\tau_i(t)} = \frac{\dot{E}_{\text{syn},i}}{E_{\text{syn},i}} - \frac{\dot{E}_{\text{drift},i}}{E_{\text{drift},i}}.$$

To account for hysteresis:

$$\dot{m}_i = \frac{\tau_i - m_i}{\tau_{h,i}}, \quad \kappa_{R,i}^{\text{eff}} = \kappa_{R,i} - h_i(\tau_i - m_i),$$

with memory timescale $\tau_{h,i} > 0$ and hysteresis gain $h_i \geq 0$.

2 Exogenous Forcing Channels

Let $\mathcal{K}$ index channels by which an external driver couples to Earth:

$$\mathcal{K} = \{\text{grav/tidal, EM/plasma, particulate/dust, radiative, informational}\}.$$

Each channel k has driver $F_k(t)$ and coupling coefficients $L_{ik}^{(\text{syn})}, L_{ik}^{(\text{drift})} \geq 0$. Transform functions ϕ_k, ψ_k map physical driver magnitudes into changes in coherence/drift energy.

3 Cross-Domain Coupling

Cross-domain transfer is modeled by:

$$W_{ij}^{(\text{syn})}, \quad W_{ij}^{(\text{drift})},$$

with nonlinear transfer functions f_{ij}, g_{ij} that allow domain j to influence domain i depending on τ_j proximity to threshold.

4 Dynamics

The minimal dynamics for domain i are:

$$\dot{E}_{\text{syn},i} = -\lambda_{s,i} E_{\text{syn},i} + \rho_{s,i} + \sum_j W_{ij}^{(\text{syn})} f_{ij}(\tau_j) + \sum_k L_{ik}^{(\text{syn})} \phi_k(F_k(t)) + \xi_{s,i}(t), \quad (1)$$

$$\dot{E}_{\text{drift},i} = -\lambda_{d,i} E_{\text{drift},i} + \rho_{d,i} + \sum_j W_{ij}^{(\text{drift})} g_{ij}(\tau_j) + \sum_k L_{ik}^{(\text{drift})} \psi_k(F_k(t)) + \xi_{d,i}(t). \quad (2)$$

Substituting into the quotient rule gives:

$$\dot{\tau}_i = \frac{\dot{E}_{\text{syn},i}}{E_{\text{drift},i}} - \tau_i \frac{\dot{E}_{\text{drift},i}}{E_{\text{drift},i}}.$$

5 Definition of Exogenic τ -Cascade

Exogenic τ -Cascade: An external driver $F_k(t)$ that, through couplings $(L^{(\cdot)}, W^{(\cdot)})$, induces effective pressure surges or threshold crossings in τ_i , potentially propagating across multiple domains.

6 Criteria (Falsifiability)

Over window $[t_0, t_1]$, a cascade exists if for some domain i :

(C1) Threshold crossing: $\exists t : \tau_i(t)$ crosses $\tau_{c,i}$,

(C2) Pressure surge: $\max_t \kappa_{R,i}^{\text{eff}}(t) \geq \kappa_i^*$,

(C3) Memory break: $\int_{t_0}^{t_1} |\tau_i(t) - m_i(t)| dt \geq \Xi_i$.

Cross-domain cascade requires $|\mathcal{S}| \geq 2$ distinct domains meeting C1C3 in the correct causal order.

7 Taxonomy of Cascade Classes

Table 1: Exogenic τ -Cascade taxonomy. Signs indicate typical priors; must be tested.

Class	Driver F_k	Channels	Sign on Syn/Drift	Timescale	Proxies
Celestial	Comet/asteroid flyby	grav/tidal, dust, rad	syn: $\pm$, drift: $+$	sdays	tidal idx, Δ
Solar	CME, geomag storm	EM, rad	syn: $\pm$, drift: $+$	mindays	Kp/Dst, Δ
Geo→Bio	Volcano, SSW	particulate, rad	syn: $-$, drift: $+$	daysyears	AOD, UV, Δ
Techno-Symb	EMP, grid shock	EM, infra	syn: $-$, drift: $+$	msdays	outages, co
Informational	Media/AI surge	symbolic influx	syn: $\pm$, drift: $+$	minweeks	traffic, eng

8 Falsifiability Protocol

- (F1) Pre-register event window, proxies, thresholds (κ_i^*, Ξ_i) , and expected lags.
- (F2) Compute τ_i and $\kappa_{R,i}^{\text{eff}}(t)$ around events.
- (F3) Use surrogate tests: time-shifted sham events, shuffled controls.
- (F4) Control multiple comparisons (FDR or Bonferroni).
- (F5) Demand doseresponse: effect size $\propto$ driver magnitude.

If no C1C3 events occur despite adequate power, declare *null* and publish bounds on couplings $L^{(\cdot)}$.

9 Simulation Role

Simulations are not proof but tools for:

- (S1) Sensitivity analysis (200500 runs per class suffices).
- (S2) Estimating plausible lag windows between domains.
- (S3) Power analysis: how many real events needed to detect effect size at chosen α .

Real falsification requires event-locked data (e.g., 510 strong solar storms, $\text{VEI} \geq 4$ eruptions). Do not simulate weak cases (ordinary asteroid flybys) before high-signal classes.

10 Limits and Non-Overreach

- Celestial flybys at safe distances are expected to be null; positive claims require extraordinary evidence.
- Solar and volcanic drivers are the strongest real test cases.
- Informational/techno-symbolic cascades are speculative but testable if preregistered.
- ENT does not assert comets change consciousness; it provides a falsifiable structure to test whether any external forcing propagates recursively into domains of interest.

11 Conclusion

The Exogenic τ -Cascade is a general, falsifiable ENT construct. It is not comet-specific but encompasses any external forcing that injects coherence or drift into recursive domains. The framework unifies diverse phenomena under one structure while demanding strict pre-registration, event-locked tests, and acceptance of null results. Thus, ENT makes the bold claim *testable*: either cascades appear with doseresponse under strong exogenic drivers, or upper bounds on couplings are established. Both outcomes advance the theory.